

Guided Notes: A Concrete Case Where We *Know* the Bayes Classifier

Learning goals

By the end of this activity, you should be able to:

- State the Bayes classification rule in terms of posterior probability.
 - Use Bayes' rule to rewrite the decision in terms of likelihood $\times$ prior.
 - Derive the Bayes classifier for a simple **two-class Gaussian** model.
 - Identify the **decision threshold** and interpret it.
 - Explain how knowing the Bayes classifier helps us benchmark ML models.
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Setup: a world where the "truth" is known

We consider a binary classification problem with:

- **Classes:** $Y \in \{0, 1\}$
- **Feature:** a single real-valued feature $X \in \mathbb{R}$

Assume the data are generated as:

Priors (class frequencies)

$$P(Y = 0) = \frac{1}{2}, \quad P(Y = 1) = \frac{1}{2}$$

Class-conditional distributions (likelihoods)

$$X \mid Y = 0 \sim \mathcal{N}(\mu_0, \sigma^2), \quad X \mid Y = 1 \sim \mathcal{N}(\mu_1, \sigma^2)$$

with $\mu_0 \neq \mu_1$ and the **same variance** σ^2 .

Interpretation: Each class generates points around its own mean. The "noise level" is σ .

1) The Bayes classifier definition

The Bayes classifier predicts the class with largest posterior probability:

$$\hat{y}(x) = \arg \max_{y \in \{0,1\}} P(Y = y \mid X = x)$$

Guided check

- What does $P(Y = y \mid X = x)$ mean in words?

- **Your answer:** _____

2) Rewrite using Bayes' rule

Bayes' rule:

$$P(Y = y \mid X = x) = \frac{p(X = x \mid Y = y) P(Y = y)}{p(X = x)}$$

Since $p(X = x)$ is the same for both classes, the argmax does not depend on it:

$$\hat{y}(x) = \arg \max_{y \in \{0,1\}} p(x \mid y) P(y)$$

Guided check

- Why can we ignore $p(x)$ in the argmax?

- **Your answer:** _____

3) Plug in our priors

Here the priors are equal:

$$P(0) = P(1) = \frac{1}{2}$$

So the decision rule simplifies to:

$$\hat{y}(x) = \arg \max_{y \in \{0,1\}} p(x \mid y)$$

Meaning: choose the class under which x is more likely.

4) Write the Gaussian likelihoods

For a normal distribution $\mathcal{N}(\mu, \sigma^2)$,

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

So:

$$p(x \mid 0) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_0)^2}{2\sigma^2}\right)$$

$$p(x \mid 1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_1)^2}{2\sigma^2}\right)$$

Guided check

- Which pieces are the same for both classes?
 - **Your answer:** _____
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5) Compare likelihoods by comparing log-likelihoods

We decide class 1 if:

$$p(x | 1) > p(x | 0)$$

Take logs (log is monotone increasing, so it preserves "greater than"):

$$\log p(x | 1) > \log p(x | 0)$$

Because the constant $-\frac{1}{2} \log(2\pi\sigma^2)$ appears on both sides, it cancels.

What remains is:

$$-\frac{(x - \mu_1)^2}{2\sigma^2} > -\frac{(x - \mu_0)^2}{2\sigma^2}$$

Multiply both sides by $-2\sigma^2$ (note the inequality flips when multiplying by a negative number):

$$(x - \mu_1)^2 < (x - \mu_0)^2$$

Interpretation: pick the class whose mean is *closer* to x .

6) Solve for the decision boundary (the threshold)

We want the set of x where both classes are equally likely:

$$(x - \mu_1)^2 = (x - \mu_0)^2$$

Expand both sides:

$$x^2 - 2\mu_1 x + \mu_1^2 = x^2 - 2\mu_0 x + \mu_0^2$$

Cancel x^2 on both sides:

$$-2\mu_1 x + \mu_1^2 = -2\mu_0 x + \mu_0^2$$

Bring the x -terms to one side:

$$2(\mu_0 - \mu_1)x = \mu_0^2 - \mu_1^2$$

Factor the right-hand side:

$$\mu_0^2 - \mu_1^2 = (\mu_0 - \mu_1)(\mu_0 + \mu_1)$$

Cancel $(\mu_0 - \mu_1)$ (valid because $\mu_0 \neq \mu_1$):

$$2x = \mu_0 + \mu_1$$

So the boundary is:

$$x^* = \frac{\mu_0 + \mu_1}{2}$$

7) Final Bayes classifier (known exactly)

Assuming $\mu_1 > \mu_0$, the Bayes classifier is:

$$\hat{y}(x) = \begin{cases} 0 & x < \frac{\mu_0 + \mu_1}{2} \\ 1 & x \geq \frac{\mu_0 + \mu_1}{2} \end{cases}$$

Guided check

- In words, how do we classify a point x ?
 - **Your answer:** _____
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8) A numeric example you can compute quickly

Let:

- $\mu_0 = 0$
- $\mu_1 = 2$
- $\sigma = 1$

Then:

$$x^* = \frac{0 + 2}{2} = 1$$

So:

- if $x < 1$, predict class 0
- if $x \geq 1$, predict class 1

Practice

Classify each point:

- $x = -0.2$: _____
 - $x = 0.9$: _____
 - $x = 1.0$: _____
 - $x = 3.4$: _____
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9) How do we *use* the fact that Bayes is known?

Use case A: Benchmarking ML models (simulation study)

If we generate training/test data from this known model, then:

- the Bayes classifier is the **best possible** classifier
- its error is the **irreducible error** for this problem

So if a learned model has test error much larger than Bayes error, the gap is due to:

- limited data (estimation error),
- model mismatch,
- optimization issues,
- etc.

Use case B: Understanding decision boundaries

This derivation shows:

- equal-variance Gaussians $\Rightarrow$ a **linear** decision rule (a threshold in 1D)
- the boundary sits exactly midway between means (when priors are equal)

Use case C: Connecting to LDA

This is the 1D version of **Linear Discriminant Analysis (LDA)**. In higher dimensions, the boundary is a hyperplane when covariances are equal.

10) Quick “what if” questions (concept checks)

1. If $P(Y = 1)$ were larger than $P(Y = 0)$, would the threshold stay at $\frac{\mu_0 + \mu_1}{2}$?
 - **Prediction:** _____
 2. If the variances were different ($\sigma_0^2 \neq \sigma_1^2$), would the boundary still be a single midpoint threshold?
 - **Prediction:** _____
 3. What happens to the overlap (and difficulty) when σ increases?
 - **Prediction:** _____
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Summary

- Bayes classifier: choose the class with highest posterior $P(y | x)$.

- With equal priors, this is equivalent to choosing the larger likelihood $p(x | y)$.
- For two Gaussians with equal variance, the Bayes decision is a simple threshold:

$$x^* = \frac{\mu_0 + \mu_1}{2}$$

- Knowing Bayes lets us quantify the *best achievable performance* and benchmark learning algorithms.